

DeepShield ApprovalGuard

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The problem

When a sensitive instruction arrives by voice or video, one combined score can conceal missing or conflicting evidence. DeepShield exposes separate media-integrity signals so a reviewer can examine the recording and decide what needs checking.

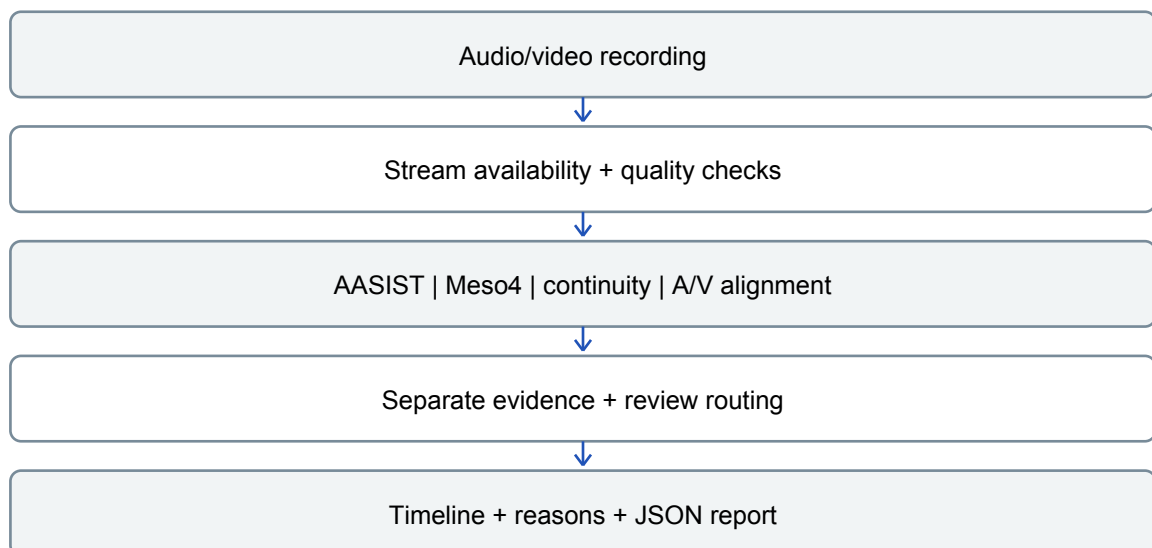
What a user gets

The workbench returns timestamped voice, face, continuity and audio/video alignment evidence, quality checks, review routing and a downloadable JSON record.

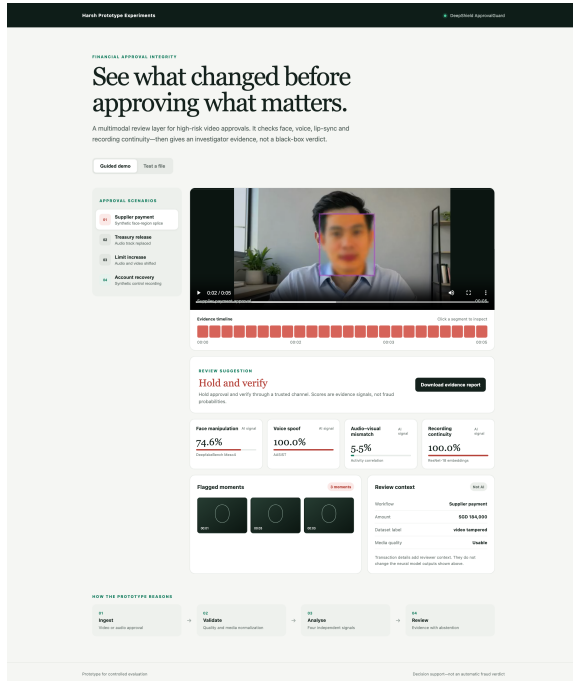
Practical value

The demonstrated workflow supports inspection of media signals. The measured benchmark covers the audio branch only, with threshold selection on the same subset. The combined workflow has not been independently validated as a real-world deepfake detector.

Logic and flow



An actual output



Saved project output from the repository. The example illustrates the workflow; benchmark results and limitations are described separately below.

Data included with the project

The repository is immediately testable with 48 labelled files:

- **24 five-second MP4 workflow fixtures:** six controls, six audio replacements, six visible identity-region edits, and six audio-video desynchronisation cases;
- **24 ASVspoof 2021 DF audio examples:** 12 bona-fide and 12 spoof clips for direct voice-model testing.

Labels, transformations, affected intervals, and provenance are documented in `dataset/manifest.csv` and `dataset/asvspoof_examples/manifest.csv`. The MP4 files are functional fixtures, not a scientific accuracy benchmark. `DATASET_CARD.md` explains the evaluation boundaries, while `DATASET_OPTIONS.md` identifies larger datasets for broader testing.

Technical snapshot

Question	Implementation
What enters the system?	Audio-only or audio-video financial instruction recordings
What is analysed?	Synthetic voice, face manipulation, visual continuity, media quality, and audio-video timing
How is evidence presented?	Separate branch scores, timestamped segments, reasons, quality fields, and a downloadable JSON record
When does it abstain?	When required streams are absent, unusable, or provide insufficient evidence
What does the routing layer produce?	STANDARD, REVIEW, ESCALATE, or INSUFFICIENT EVIDENCE
What does it not claim?	Identity verification, intent inference, a calibrated fraud probability, or production readiness

How it works

AI components

Evidence branch	Implementation	What it contributes
Synthetic voice	AASIST spectro-temporal graph-attention network	Spoof evidence for the complete clip and temporal windows
Face manipulation	DeepfakeBench Meso4 compact CNN	Frame-level face-region manipulation evidence
Recording continuity	ImageNet-pretrained ResNet-18 embeddings	Learned visual discontinuity between neighbouring segments
Audio-video timing	Mouth-motion and audio-activity correlation	Interpretable synchronisation evidence; this branch is signal processing rather than a trained lip-reading model

The decision layer deliberately keeps these signals separate. Its priority index is a routing aid, not a fraud probability. Two high signals suggest escalation; one high signal or two elevated signals suggest independent review. Poor-quality or missing inputs can produce an insufficient-evidence result.

Locally measured voice-model result

AASIST was evaluated locally on a balanced 570-file subset of ASVspoof 2021 DF:

Metric	Result
ROC-AUC	0.9078
Equal error rate	18.60%
Accuracy at the subset-selected EER threshold	81.40%
Spoof precision	81.40%
Spoof recall	81.40%
Mean model inference time on Apple MPS	17.44 ms/file

The threshold was selected and measured on the same showcase subset, so these numbers describe the implementation experiment rather than an independent production estimate. The exact summary and all 570 per-file scores are available in [evaluation/](#).

Evaluation status

The saved quantitative experiment evaluates the **AASIST audio branch only**. The visual, continuity, synchronisation, and combined routing branches are demonstrated with functional fixtures, not validated as a complete multimodal detector on an independent held-out dataset. The 24 MP4 cases verify workflow behaviour under known transformations; they do not establish generalisation to real attacks.

Technical boundaries

- The implementation detects media-integrity evidence; it does not establish identity or intent.
- AASIST was trained in an audio anti-spoofing domain and can shift under unseen speakers, codecs, noise, and generators.
- Meso4 uses a portable centre-region crop here; a production pipeline would require robust face detection and alignment.
- The demonstration videos are workflow fixtures outside the training domains of AASIST and Meso4.
- Scores are evidence scales, not calibrated probabilities.
- Deployment would require representative institutional data, independent validation, threshold calibration, monitoring, access controls, and human review.

See `THIRD_PARTY_MODELS.md` for model provenance and checkpoint hashes, and `API.md` for programmatic use.

Evidence and reproduction references

Source revision: `de51602f65cb9a7e669845da193a3d46240b53ef`

- [README.md](#)
- [docs/output-example.json](#)

This report describes the source and saved evidence at the revision above. Training and full benchmark runs were not repeated for this documentation release. Dataset, model and dependency licences remain separate from the project documentation.